

Lethukuthula Nothando Mthiyane

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OBJECTIVE

To join a Graduate position where I can apply my hands-on experience in data engineering, data analysis and Python to drive business strategy and deliver meaningful insights through data-driven solutions. With a BSc in Computer Science and Physics, an ongoing BSc Honours in Computing at UNISA and real industry experience contributing to the AIOPs Single Pane of Glass Dashboard at BCX, I am committed to growing within a forward-thinking analytics team that values technical excellence, continuous learning and making an impact that matters.

EDUCATION

BSc Honours in Computing (Computer Science & Information Systems)

University of South Africa (UNISA) | Feb 2025 - Present

Modules: Software Engineering, Machine Learning, NLP, Information Security, Project Management

Bachelor of Science: Computer Science and Physics (Double Major)

University of Zululand, South Africa | Feb 2016 – Nov 2019

- **Computer Science:** Advanced Programming, Software Development, Distributed Systems, Networking
- **Mathematics:** Calculus, Advanced Calculus, Linear Mathematics, Discrete Mathematics
- **Physics:** Electronics, Nuclear Physics, Mechanics, Quantum Physics

National Certificate: Information Technology (Systems Development)

Specialization: Data Science

Explore AI Academy, South Africa | Jan 2022 – Nov 2022

National Certificate: Grade 12

Aquadene Secondary School, South Africa | Jan 2009 – Nov 2013

Stream: Mathematics and Science

WORK EXPERIENCE

Data Engineering Learner | CAPACITI Digital Career Accelerator | Johannesburg

Nov 2025 – Present

- Completed AI Bootcamp earning a verified credential via Coursera
- Achieved Data Engineer Practitioner certification under the SFIA Framework
- Completed IBM certifications in Data Engineering and Python
- Building hands-on skills in data pipelines, ETL processes, cloud platforms and big data tooling
- Applying Python and SQL to real-world datasets through Agile sprints and technical assessment

Tech Officer: Call Desk Agent | BCX | Remote

Jun 2023 - May 2024

- Provided Tier-1 technical support using remote diagnostic tools to resolve system anomalies
- Monitored live dashboards conducting data refreshes to detect and report outliers immediately
- Generated and distributed monthly operational reports to relevant stakeholders

Data Scientist Intern | BCX | Remote

Sep 2022 – Dec 2022

- Contributed to the end-to-end engineering of the AIOPs Single Pane of Glass Dashboard, a real-time IT operations monitoring solution used across BCX's operational environment

- Applied ETL techniques to process Big Data from the Cloudera Data Platform enabling downstream analytics and ML workloads
- Used Python, SQL, Apache Spark and Hue to clean, transform and prepare complex unstructured operational datasets
- Supported MLOps workflows contributing to model preparation pipelines feeding predictive models into the dashboard
- Awarded Certificate of Contribution by BCX for impact delivered during internship

Data Science Trainee | Explore AI Academy | Remote

Jan 2022 – Aug 2022

- Built a regression model to forecast energy demand shortfalls achieving 90% model accuracy
- Developed NLP classification models to analyse and predict user sentiment from text data
- Engineered a collaborative filtering movie recommendation system using unsupervised learning

PROJECTS

DEEPA | CAPACITI AI Bootcamp | 2025

AI-powered marketing tool that generates ad scripts, social media posts and branded content using Generative AI and NLP. Designed to democratise professional marketing for small businesses.

Live: [DEEPA](#) | **Tools:** React | Node.js | Generative AI | NLP

DermaScan AI | CAPACITI AI Bootcamp | 2025

Skin health assistant detecting potentially malignant moles using Computer Vision making dermatological screening accessible to underserved communities.

Live: [Dermascan AI](#) | **Tools:** Python | TensorFlow | Flask | Computer Vision

Unibridge | CAPACITI AI Bootcamp | 2025

Accessibility platform for vision impaired, deaf and speech impaired users with real-time text-to-speech, speech-to-text and translation.

Live: [Unibridge](#) | **Tools:** JavaScript | React | Speech Recognition | NLP

Space Portfolio Website | Personal | 2025

Personal portfolio with 3D animations built on Next.js 14 deployed on Vercel.

Live: [lethukuthula-portfolio.vercel.app](#) | **Tools:** Next.js | TypeScript | Tailwind CSS | Framer Motion

TECHNICAL SKILLS

Data Engineering: ETL/ELT Pipelines | Apache Spark | Cloudera | Hadoop | Hive | Big Data Processing

Programming: Python (Pandas, NumPy, Scikit-learn) | SQL | Java | JavaScript

Data Science and AI: Machine Learning | NLP | Generative AI | Computer Vision | MLOps | EDA

Visualisation: Power BI | Microsoft Excel | matplotlib | Dashboard Development

Platforms: IBM Cloud | Cloudera Workbench | Vercel | Jupyter Notebook

Soft Skills: Agile Methodology | Problem Solving | Design Thinking | Teamwork

CERTIFICATIONS

- **Data Engineer Practitioner SFIA Framework** (Coursera, 2025)
- **Artificial Intelligence Bootcamp** (Coursera, 2025)
- **Data Analytics** (Google x Coursera)